

14-2c The Firm's Short-Run Decision to Shut Down

So far, we have been analyzing the question of how much a competitive firm will produce. In some circumstances, however, the firm will decide to shut down and not produce anything at all.

Here we need to distinguish between a temporary shutdown of a firm and the permanent exit of a firm from the market. A *shutdown* refers to a short-run decision not to produce anything during a specific period of time because of current market conditions. *Exit* refers to a long-run decision to leave the market. The short-run and long-run decisions differ because most firms cannot avoid their fixed costs in the short run but can do so in the long run. That is, a firm that shuts down temporarily still has to pay its fixed costs, whereas a firm that exits the market does not have to pay any costs at all, fixed or variable.

For example, consider the production decision that a farmer faces. The cost of the land is one of the farmer's fixed costs. If the farmer decides not to produce any crops one season, the land lies fallow, and he cannot recover this cost. When making the short-run decision of whether to shut down for a season, the fixed cost of land is said to be a *sunk cost*. By contrast, if the farmer decides to leave farming altogether, he can sell the land. When making the long-run decision of whether to exit the market, the cost of land is not sunk. (We return to the issue of sunk costs shortly.)

Now consider what determines a firm's shutdown decision. If the firm shuts down, it loses all revenue from the sale of its product. At the same time, it saves the variable costs of making its product (but must still pay the fixed costs). Thus, *the firm shuts down if the revenue that it would earn from producing is less than its variable costs of production*.

A bit of mathematics can make this shutdown rule more useful. If TR stands for total revenue and VC stands for variable cost, then the firm's decision can be written as

$$\text{Shut down if } TR < VC.$$

The firm shuts down if total revenue is less than variable cost. By dividing both sides of this inequality by the quantity Q , we can write it as

$$\text{Shut down if } TR/Q < VC/Q.$$

The left side of the inequality, TR/Q , is total revenue $P \times Q$ divided by quantity Q , which is average revenue, most simply expressed as the good's price, P . The right side of the inequality, VC/Q , is average variable cost, AVC . Therefore, the firm's shutdown rule can be restated as

$$\text{Shut down if } P < AVC.$$

That is, a firm chooses to shut down if the price of the good is less than the average variable cost of production. This rule is intuitive: When choosing whether to produce, the firm compares the price it receives for the typical unit to the average variable cost that it must incur to produce the typical unit. If the price doesn't cover the average variable cost, the firm is better off stopping production altogether. The firm still loses money (because it has to pay fixed costs), but it would lose even more money by staying open. The firm can reopen in the future if conditions change so that price exceeds average variable cost.

We now have a full description of a competitive firm's profit-maximizing strategy. If the firm produces anything, it produces the quantity at which marginal cost equals the good's price, which the firm takes as given. Yet if the price is less than average variable cost at that quantity, the firm is better off shutting down temporarily and not producing anything. These results are illustrated in [Figure 3](#). *The competitive firm's short-run supply curve is the portion of its marginal-cost curve that lies above the average-variable-cost curve.*

Figure 3

The Competitive Firm's Short-Run Supply Curve

In the short run, the competitive firm's supply curve is the portion of its marginal-cost curve (MC) that lies above its average-variable-cost curve (AVC). If the price falls below average variable cost, the firm is better off shutting down temporarily.